Simple transfinite induction: a formula holding at zero, closed under successors, and closed under limits with prior instances holds for all ordinals

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the three enumerated hypotheses, the least-counterexample sentence and the case division are verbatim in the span; each case's clash with its hypothesis is one-line immediate from the statement's display.

Editorial note. The label simpletransrecursion names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Transfinite induction / least counterexample. Same unit, \ollabel{ordinductionschema} (source line 446). Supplies the least ordinal gamma falsifying phi that the contrapositive proof argues about.
- Successor and limit ordinal trichotomy for ordinals. The unit's development of successor and limit ordinals in the preceding section (the case split at lines 947-951 uses exactly this division). Every ordinal is zero, a successor, or a limit, so a least counterexample must fall in one of the three hypotheses.

Theorem 0.1 (Simple Transfinite Induction)**.** *Let $\phi(x)$ be a formula such that:*

1. *$\phi(\emptyset)$; and*
2. *for any ordinal α, if $\phi(\alpha)$ then $\phi(+1\alpha)$; and*
3. *if α is a limit ordinal and $(\forall \beta \in \alpha)\phi(\beta)$, then $\phi(\alpha)$.*

Then $\forall \alpha \phi(\alpha)$.

Proof. We prove the contrapositive. So, suppose there is some ordinal which is $\neg\phi$; let γ be the least such ordinal. Then either $\gamma = \emptyset$, or $\gamma = +1\alpha$ for some α such that $\phi(\alpha)$; or γ is a limit ordinal and $(\forall\beta \in \gamma)\phi(\beta)$. □